All atoms of a specific element have exactly the same chemical makeup, size, and mass. Each atom has an atomic number, which represents the number of protons that are in the nucleus of a single atom of that element. There are a total of 118 elements. Many elements occur naturally on Earth; however, some are created in laboratory by scientists.

9.1.1 Symbols of Elements

A symbol is an image, object, etc., that stands for some meaning. For instance, a dove is a symbol of peace. Similarly, we denote mathematical operations by symbols. For example '+' denotes addition; '–' denotes subtraction etc. In the same way in chemistry each element is denoted by a symbol. Writing out the name of an element every time would become too troublesome. So, the name of an element is represented by shortened form called as symbol. Let us learn the brief history of symbols of elements.

a. Greek Symbols

The symbols in the form of geometrical shapes were used by the ancient Greeks to represent the four basic elements around us such as earth, air, fire and water.

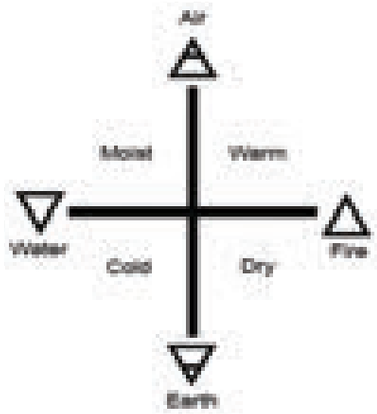


Figure 9.1 Greek symbols

b. Alchemist Symbols

In the days of alchemists, different materials that people used were represented by different symbols while they tried to change less valuable metal into gold. That process was called **alchemy** and the men who did this work were known as **alchemists**.



Nickel *Arsenic* *Antimony* *Water*

Figure 9.2 Symbols used by Alchemist

c. Dalton Symbols

In 1808, John Dalton, English scientist tried to name various elements based on pictorial symbols. Those symbols are difficult to draw and hence they are not used. It is only of historical importance.

	Hydrogen		Copper
	Nitrogen		Lead
	Carbon		Water
	Sulphur		Ammonia
	Phosphorus		Olefiant
	Alumina		Carbonic Oxide
	Soda		Carbonic Acid
	Pot Ash		Sulphuric Acid
	Oxygen		

Figure 9.3 Dalton Symbols

d. Berzelius Symbols

In 1813, Jon Jakob Berzelius devised a system using letters of alphabet rather than signs. The modified version of Berzelius system follows under the heading 'System for Determining Symbols of the Elements'.

e. Present system for determining symbols of the elements

1. The symbols of the most common elements, mainly non-metals, use the first letter of their English name.

Table 9.1 Elements having first letter as symbol

Element	Symbol	Element	Symbol
Boron	B	Oxygen	O
Carbon	C	Phosphorus	P
Fluorine	F	Sulphur	S
Hydrogen	H	Vanadium	V
Iodine	I	Uranium	U
Nitrogen	N	Yttrium	Y

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